

Testing a planar lattice-optimality theorem against an exciton-fluid interaction: kernel-checked lemmas, a four-flavour model theorem, and N-particle tests on rusty-SUNDIALS

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<https://github.com/xaviercallens/SocrateAI-Scientific-QuantumFluids>

October 2026

Abstract

A planar universal-optimality theorem for completely monotone pair interactions (OpenAI, September 2026: AI-generated and not yet refereed; formalised in Lean, and accepted by the Comparator tool in a run by the author's own LeanMaster project, with no human audit of the statement) says that the triangular lattice minimises the energy per particle among all locally finite planar configurations of centred density one. We ask, as a dependency check, what it says about the interaction between the dipolar interlayer excitons of a 2026 equilibrium exciton-condensate experiment. In Lean 4, with standard axioms only, we prove that differencing preserves complete monotonicity, so that the direct exciton–exciton kernel built from *any* admissible single-layer potential is covered; the finite telescoping identity behind the capacitor formula for the Hartree coefficient (the factor π of the polar integral and the limit are prose steps); that the uniform density minimises a positive-semidefinite translation-invariant interaction; and, conditionally on the theorem taken as a hypothesis, the periodic-extension step that turns it into a bound for N particles on a torus. The consequence is a classical lower bound on the energy per particle which, at the experimental densities, lies 8–22 times below the Hartree energy: the sandwich is wide, and the bound is structural, not predictive. We prove six statements about the four-flavour mean-field model of the experiment and reproduce T1–T3 and T5 with CVODE flows (T2 and T3 to 1.68×10^{-12}). We then test the theorem numerically, in a design whose gates were fixed before the runs, with gradient flows of $N = 36$ and 64 particles in tori commensurate and incommensurate with the lattice, integrated by rusty-SUNDIALS CVODE, with non-completely-monotone and planted-bug controls. In 2060 gradient-flow runs of completely monotone kernels on tori of $N = 36$ and 64 the bound was never violated (smallest $E/e_{\text{lat}} - 1 = -1.1 \times 10^{-13}$, at the roundoff and truncation level of the lattice sums; registered tolerance 10^{-10}); the lattice energy was attained on 8 of 9 commensurate cases as registered (the ninth only by a longer exploratory flow); the best energies found on incommensurate tori exceed it by 7.9×10^{-7} to 8.6×10^{-3} ; and the non-completely-monotone control (800 runs) goes far below it (best $E/e_{\text{lat}} = 0.663\text{--}0.724$). A mean-field test with a nonlocal kernel passes four of its five registered gates. We report what went wrong: a lattice-enumeration slip shared by two implementations and caught only by a closed-form constant; a compute-budget amendment; a registered stopping rule too short for soft modes; and a control kernel of the mean-field test whose grand-canonical functional is unbounded below (that gate failed as registered, and the control had been changed from its committed depth before the run). Producer and verifiers of the Lean files are instances of the same model; nothing here has been refereed.

1 Introduction

In 2026 Qi *et al.* reported evidence for two-component Bose–Einstein condensates of dipolar interlayer excitons in a MoSe₂/hBN/WSe₂ electron–hole bilayer with an interlayer distance $d \approx 2$ nm [1]. The excitons are injected electrically and are in equilibrium; there are four spin–valley flavours; the evidence is the spin–valley susceptibility, read through a four-variable mean-field model; the condensate signatures persist up to about 1.8 K (in version 1 the Berezinskii–Kosterlitz–Thouless temperature reaches that value near the exciton Mott density $n \approx 0.75 \times 10^{12} \text{ cm}^{-2}$). The authors state that counterflow transport, the hallmark of exciton superfluidity, is experimentally inaccessible in current TMD bilayers (version 1, Introduction).

In September 2026 OpenAI released a proof, with a Lean formalisation, that the triangular lattice minimises the energy per particle among *all* locally finite planar configurations of centred density one for *every* smooth completely monotone function of the squared distance [2]. This is the planar case of the Cohn–Kumar universal-optimality conjecture [3], whose dimensions 8 and 24 were settled in 2022 [4].

Dipolar excitons repel each other; the interaction is a function of the distance. The question of this note is therefore a *dependency check*: which hypotheses of the theorem does the exciton physics meet, which consequences follow, and how large are they? Our answers are given in Sections 3 to 4; their kernel-checked parts are listed in Section 5, and numerical tests with rusty-SUNDIALS, a Rust implementation of the CVODE algorithms of the SUNDIALS suite [5], are in Section 6.

What this note claims and what it does not

Claims. (i) A lemma, proved in Lean, that makes the theorem applicable to the exciton–exciton direct kernel; (ii) the size of the resulting lower bound relative to the Hartree upper bound at the experimental densities; (iii) six statements about the experiment’s own model, proved in Lean (T1–T3 and T5 also reproduced by CVODE flows); (iv) the periodic-extension step that turns the theorem into a bound for N particles on a torus, proved in Lean as a statement conditional on the theorem; (v) pre-registered numerical tests of the theorem on small tori, with controls, and of the uniform mean-field state with a nonlocal kernel.

Does not claim. The theorem is AI-generated and unrefereed; we use it as a hypothesis and test it only on finite systems, which can fail to falsify it but cannot prove it. The Lean files were produced and checked by instances of the same model (producer $\neq$ verifier, not human review). Nothing here describes the experiment beyond the authors’ own model, whose parameters g_X and Δ are phenomenological (and differ between version 1 and the version of record); agreement of magnitude with the data is not independent evidence. The sandwich between the lattice and Hartree energies is wide in the experimental window (a factor 8–22): the ground-state energy lies between them and nothing here locates it. No claim is made about the solver being the best of its kind. Neither the experimental group nor the authors of the theorem have reviewed this note. It was produced overnight by an AI system (Claude, Anthropic) working autonomously under the author’s standing instructions, with two further instances as verifiers and one as adversarial reviewer; no human reviewed it before its deposit.

2 The theorem and its conventions

Let $g: (0, \infty) \rightarrow [0, \infty)$ be smooth with $(-1)^r g^{(r)}(t) \geq 0$ for all $r \geq 0$, $t > 0$ (“admissible”, $g \in \text{Adm}$). For a locally finite $C \subset \mathbb{R}^2$ write $N_R = \#(C \cap B_R)$, with B_R the closed disk about the origin, and

$$E_g(C) = \liminf_{R \rightarrow \infty} \frac{1}{N_R} \sum_{\substack{x, y \in C \cap B_R \\ x \neq y}} g(|x - y|^2), \quad (1)$$

the sum running over *ordered* pairs. If $N_R/(\pi R^2) \rightarrow 1$ (centred density one), the theorem states $E_g(C) \geq \sum_{a \in A \setminus 0} g(|a|^2) = E_g(A)$, with A the triangular lattice of covolume one, in $[0, \infty]$. Four features matter for every use.

1. *Ordered pairs.* The physical energy per particle is half of E_g : $e_{\text{lat}}(\rho) = \frac{1}{2} \sum_{a \in A_\rho \setminus 0} V(|a|)$, with $A_\rho = \rho^{-1/2} A$. A factor-two slip is the likeliest error downstream; Section 6 plants one on purpose.
2. *Centred disks and a lim inf.* The statement is about infinite configurations. A bound for N particles in a torus needs a periodic-extension argument (Proposition 1), which brings in the interaction of a particle with its own periodic images; that step is formalised here, conditionally on the theorem.
3. *Value, not minimisers.* The minimiser is not unique in this formulation: a single vacancy changes neither the density nor the lim inf. Uniqueness of the triangular lattice is not part of the theorem.
4. *Class, not each potential.* The proof constructs sharp Fourier minorants for every Gaussian and integrates; a sharp auxiliary function for each completely monotone potential is not claimed [2].

The Lean statement is `OAI.AtomicTriangular.universal_energy_minimum`; the Comparator tool accepted its proof with axioms `{propext, Quot.sound, Classical.choice}` in a 140-minute run of the author’s own LeanMaster project on a second machine [6] (the external kernel nanoda was not run; the statement has not been audited by a human; a statement-fidelity audit against Definitions 1.1–1.3 of [4] exists, written by the AI agents of that project and therefore not independent). That check was not repeated in this study. Predecessors are Montgomery’s theorem among lattices [7] and the dimension-8 and 24 results [4].

3 Kernels, a lemma, and a lower bound

The lemma. Two interlayer excitons, each with its electron in one layer and its hole in the other at distance d , interact directly through $V_{xx}(r) = 2[V(r) - V(\sqrt{r^2 + d^2})]$ in terms of the single-layer electron–electron potential V (the electron–hole term is the second). This treats the excitons as point dipoles (the in-plane size of an exciton is neglected) and uses the same V for the intra- and interlayer terms. In the variable $t = r^2$ it is $2[g(t) - g(t + d^2)]$.

Lemma 1 (differencing). *If $g \in \text{Adm}$ and $c > 0$ then $t \mapsto g(t) - g(t + c) \in \text{Adm}$.*

Proof. Put $h_r = (-1)^r g^{(r)} \geq 0$. Then $h'_r = -h_{r+1} \leq 0$, so each h_r is non-increasing. Hence $(-1)^r (g - g(\cdot + c))^{(r)}(t) = h_r(t) - h_r(t + c) \geq 0$ for all $r \geq 0$ (for $r = 0$ this is non-negativity). $\square$

The statement is classical (a forward difference of a completely monotone function is completely monotone, immediately from the Bernstein representation); the point of the Lean proof is that it needs nothing beyond the definition and is mechanical. With $g(t) = t^{-1/2}$ (the Coulomb case of the Riesz family, admissible by an explicit formula for the derivatives) the lemma gives the bilayer dipole kernel $2[t^{-1/2} - (t + d^2)^{-1/2}]$, which decays as $d^2 t^{-3/2}$ and has a finite lattice energy. The same lemma covers every exciton–exciton kernel built from an admissible single-layer potential, in particular the Keldysh–Rytova potential, which is a positive mixture of exponentials $e^{-ur/r_0}/\sqrt{1+u^2}$ by the integral representation of $H_0 - Y_0$ [8] (checked numerically at three points to better than 2×10^{-31} by a scratch script kept in the repository, not proved in Lean); each exponential $e^{-c\sqrt{t}}$ is completely monotone in $t = r^2$ (classical: $\sqrt{t}$ is a Bernstein function; not formalised); and a Yukawa potential. The potential of a dual-gated device is a sum of image charges with alternating signs; its complete monotonicity is *not* established here: we checked the sign pattern of the derivatives to order 5 at seven values of t for gate distances 5, 7.5 and 10 nm and $d = 2$ nm, which is evidence and not a proof.

Kernel-checked. In Lean 4 (Section 5): Lemma 1; that the bilayer kernel is admissible (the Riesz case $s = 1$, $t \mapsto t^{-1/2}$, is proved from Mathlib alone in a probe written by the verifying instance and integrated into the file; the general- s statement is in the LeanMaster contribution [6]); and the negative control that e^{-t^2} (the cluster-crystal kernel of the generalised exponential model) is *not* admissible.

The Hartree coefficient. For admissible g with $g \rightarrow 0$ at infinity,

$$\int_{\mathbb{R}^2} [g(|x|^2) - g(|x|^2 + d^2)] d^2x = \pi \int_0^{d^2} g(t) dt \quad (2)$$

(telescoping: polar coordinates give the factor π and the telescoping sum the limit). For $g = 2t^{-1/2}$ this is $4\pi d$, in units $e^2/4\pi\epsilon_0\epsilon = 1$: the capacitor formula $\tilde{U}(0) = e^2d/\epsilon_0\epsilon$, and the authors' $g_H = 8\pi d$ in Rydberg units. What Lean proves is the finite identity $\int_0^R [g(t) - g(t+c)] dt = \int_0^c g - \int_R^{R+c} g$ (`telescoping_integral`) and $\int_0^{d^2} 2t^{-1/2} dt = 4d$ (`bilayer_hartree`); the factor π and the limit $g \rightarrow 0$ are prose steps.

Mean field. A completely monotone function of r^2 is positive definite in every dimension [9], so its Fourier transform is non-negative. For a positive-semidefinite translation-invariant interaction on a finite abelian group, the uniform density minimises $\sum_{x,y} n(x)n(y)U(x-y)$ at fixed mass (Lean: `uniform_minimises`, with positive semidefiniteness as a hypothesis). Consequently a Gross–Pitaevskii calculation with such a kernel cannot crystallise: the uniform state is its ground state, and the Bogoliubov frequencies are real ($\omega_k^2 > 0$ for $k \neq 0$): no dynamical instability and no roton softening (a maxon–roton structure of ω_k is not excluded). That is a statement about mean-field theory, not about the quantum fluid.

Proposition 1 (sandwich). *Let N bosons live on a torus $\mathbb{R}^2/\Lambda$ with $\text{covol}(\Lambda)\rho = N$, with $H = -\frac{1}{2}\Delta + \sum_{i<j} U_\Lambda(x_i - x_j)$ and U_Λ the periodised completely monotone kernel, and let $c_\Lambda = \frac{1}{2} \sum_{m \in \Lambda \setminus 0} U(|m|)$ be the interaction of a particle with its own periodic images. Then the ground-state energy per particle satisfies $e_{\text{lat}}(\rho) - c_\Lambda \leq e_0(\rho) \leq \frac{1}{2}\rho\tilde{U}(0)(1 - 1/N) \leq e_H(\rho)$, $e_H = \frac{1}{2}\rho\tilde{U}(0)$. For the Hamiltonian $H + Nc_\Lambda$ (the energy of the periodic extension, which is what the theorem and `torus_lower_bound` bound and what Section 6 computes) the lower bound reads $e_{\text{lat}}(\rho) \leq e_0$. For kernels that decay faster than r^{-2} , $c_\Lambda \rightarrow 0$ as $N \rightarrow \infty$ at fixed ρ , so Table 1 is unaffected.*

Sketch. Upper bound: the constant orbital (a variational state, kinetic energy zero) has energy per particle $\frac{1}{2}\rho\tilde{U}(0)(1 - 1/N)$ for a non-negative kernel. *Lower bound:* the kinetic energy is non-negative, so e_0 is at least the minimum of the potential energy over N -point configurations; the periodic extension of a configuration has centred density ρ , and the theorem bounds its energy per particle, self-images included, by $e_{\text{lat}}(\rho)$; the energy per particle of the extension is the physical one plus c_Λ . The periodic-extension step (the class counts of the translates of Λ in disks, and the comparison of the disk sums with the per-cell sum) is formalised in Lean as `torus_lower_bound`, conditional on the first half of the upstream theorem, taken as a hypothesis; the integration of the pointwise bound over a probability density is elementary and is not formalised. Configurations with two particles at the same point of the torus are excluded by hypothesis (a set of measure zero for any L^2 state). $\square$

ρd^2	10^{-3}	10^{-2}	0.1	0.3	1	3
e_H/e_{lat}	44.7	14.2	4.64	2.87	1.87	1.44
n (10^{12} cm^{-2}), $d = 2 \text{ nm}$			0.1	0.3	0.5	0.75
e_H/e_{lat}			22.4	13.0	10.1	8.26

Table 1: Hartree energy over classical lattice energy for the undamped bilayer kernel (independent numpy and Rust sums agree to 9×10^{-14}). In the experimental window the sandwich is wide, a factor 8–22; it narrows towards higher density (1.44 at $\rho d^2 = 3$). Nothing is said about where the true energy lies.

4 The experiment’s own model

With flavours 1, 2 = KK, K’K’ (intravalley) and 3, 4 = KK’, K’K (intervalley), the authors’ energy is

$$H(n) = \sum_i (E_i - \mu) n_i + \frac{1}{2} (g_H + g_X) \left(\sum_i n_i \right)^2 - g_X (n_1 n_2 + n_3 n_4), \quad n_i \geq 0, \quad (3)$$

with $E_{1,2} = \pm(g_v - g_c)\mu_B B - \Delta$, $E_{3,4} = \mp(g_c + g_v)\mu_B B$, $g_H = 8\pi d$, $g_c \approx 3$, $g_v \approx 6$ and the phenomenological $g_X = 1$, $\Delta = 1 \text{ } \mu\text{eV}$ of arXiv version 1 (Rydberg and Bohr-radius units; the version of record uses $g_X = 1.5$, $\Delta = 0.3 \text{ } \mu\text{eV}$, see below; the paper’s Eqs. (2)–(3) are quoted as printed in version 1, the main text of the version of record being behind a paywall). A four-variable quadratic energy is exactly what a proof assistant handles. We proved, in Lean (Section 5):

- T1.** for $g_X > 0$ and $E_i \neq E_j$ across the two pairs, a global minimiser on the orthant has its support inside one exchange pair, $\{1, 2\}$ or $\{3, 4\}$;
- T2.** at any stationary point supported on $\{1, 2\}$: $n_2 - n_1 = 2(g_v - g_c)\mu_B B/g_X$ and $N = 2(\mu + \Delta)/(2g_H + g_X)$;
- T3.** at any stationary point supported on $\{3, 4\}$: $n_3 - n_4 = 2(g_c + g_v)\mu_B B/g_X$ and $N = 2\mu/(2g_H + g_X)$;
- T4.** on each pair subspace $H = \Omega + \frac{1}{4}(2g_H + g_X)(N - N_*)^2 + \frac{1}{4}g_X(\delta - \delta_*)^2$, where N, δ are the sum and difference of the pair’s occupations ($n_1 \pm n_2$, resp. $n_3 \pm n_4$) and N_*, δ_* the stationary values of T2 and T3, with $\Omega_A = -(\mu + \Delta)^2/(2g_H + g_X) - \beta_A^2/g_X$, $\Omega_B = -\mu^2/(2g_H + g_X) - \beta_B^2/g_X$, $\beta_A = (g_v - g_c)\mu_B B$, $\beta_B = (g_c + g_v)\mu_B B$, so that for $g_X > 0$, $2g_H + g_X > 0$ these are the minima of H on the pair subspaces;
- T5.** $\Omega_A = \Omega_B$ iff $(\mu_B B)^2 = g_X \Delta(2\mu + \Delta)/(4g_c g_v(2g_H + g_X))$ (the first-order transition; equality only: that Π_A is the ground state below B_c is not proved);
- T6.** for $g_X < 0$ and $E_i \neq E_j$ every minimiser is single-component (the Hartree–Fock ferromagnet; the negative control).

T2 and T3 are the authors’ printed formulas (a confirmation). The paper prints only qualitative forms of T1 (“no more than two flavors condense simultaneously”) and T6 (“negative g_X leads to a ferromagnetic single-component condensate”); the exchange-pair structure, the identity of T4 and the closed form of T5 are not printed. The genericity hypothesis $E_i \neq E_j$ fails on the four lines $\Delta = \pm 2g_v \mu_B B$, $\pm 2g_c \mu_B B$ of the (B, Δ) plane ($B \approx 1.4$ and 2.9 mT for the quoted values). It cannot be dropped from T1 and T6 as stated (the Lean file gives $E = (0, 10, 0, 10)$, with a minimiser spanning both pairs); on the four physical lines the conclusion appears to persist numerically but is not proved.

Critical field. With the version-1 values ($g_X = 1$, $\Delta = 1 \text{ } \mu\text{eV}$; $a_B = 1.5 \text{ nm}$, $\text{Ry} = 67 \text{ meV}$) the closed form gives $B_c = 55.9 \text{ mT}$ at $n_x = 0.5 \times 10^{12} \text{ cm}^{-2}$ (50.0 mT at 0.4×10^{12} ; their Extended Data Fig. 7c reads “ $\sim 50 \text{ mT}$ ”). The version of record states $g_X = 1.5$ and $\Delta = 0.3 \text{ } \mu\text{eV}$ as “the parameters used in the main text” (Extended Data Fig. 8: $n_x = 0.4 \times 10^{12} \text{ cm}^{-2}$, energy crossing at $\sim 30 \text{ mT}$); with them the closed form gives 33.5 mT at 0.4×10^{12} and 37.5 mT at

0.5×10^{12} (script `exploration/exciton/python/critical_field_versions.py`; $\pm 3\%$ with the unit constants of the paper’s Methods). That the order of magnitude agrees is expected, since g_X and Δ were chosen to describe the data; the experimental field windows of their Fig. 4 are averaging windows, not boundaries. The statement $B_c \propto \sqrt{n}$ at fixed Δ reproduces the trend they report.

One remark on the source, for readers who check against the paper: in arXiv version 1 the Ginzburg–Landau line of the Methods carries $-(g_v - g_c)\mu_B B(n_1 - n_2)$ and states $n_1 - n_2 = 2(g_v - g_c)\mu_B B/g_X$, the opposite sign to the paper’s Eq. (3) and to its main text ($n_2 - n_1 = \dots$ for II_A); the Lean statements follow the paper’s Eqs. (2)–(3) and its main text. This concerns version 1 only; the main text of the version of record (paywalled) was not read, only its public abstract and Extended Data legends.

5 Lean formalisation

The files are Lean 4 [10] with Mathlib [11].

file	role	named results
<code>ExcitonX1.lean</code>	Lemma 1	<code>admissible_sub_shift</code> , <code>admissible_const_mul</code> , <code>bilayerDipole_admissible</code> (given $s = 1$), <code>gem4_not_admissible</code>
<code>FourFlavour.lean</code>	T1–T6	<code>support_in_one_pair</code> , <code>IIA_polarisation</code> , <code>IIB_polarisation</code> , <code>intravalley_density</code> , <code>grand_potential_gap</code> , <code>critical_field</code> , <code>single_component_of_neg_gX</code>
<code>MeanField.lean</code>	Eq. (2), mean field	<code>uniform_minimises</code> , <code>telescoping_integral</code> , <code>bilayer_hartree</code> , <code>variational_lower_bound</code>
<code>TorusBound.lean</code>	periodic-extension step (X4), conditional on the upstream theorem as a hypothesis	<code>periodic_universal_lower_bound</code> , <code>torus_lower_bound_unit</code> , <code>torus_lower_bound</code> (density ρ), <code>torus_energy_per_particle</code> , <code>admissible_comp_div</code> ; verbatim copy of upstream’s definitions
<code>GradientFlow.lean</code>	energy monitor	<code>energy_antitone</code> , <code>energy_monotone_ascent</code> (control)
<code>FourFlavourNegativeControl.lean</code>	control	wrong-sign polarisation, <i>fails to compile by design</i>

Table 2: The Lean files (`exploration/exciton/lean/`). Every named result depends only on `{propext, Classical.choice, Quot.sound}`; no `sorry`. The files as deposited were compiled under Lean 4.34.1 with Mathlib tag v4.34.1 (commit d13f23b7) (the pin of the upstream project); all but `TorusBound.lean` were also compiled, as deposited, under Lean 4.34.0-rc2 / Mathlib 85e3a25 (the library’s pin until 2026-10-11).

Verification status. Producer and verifiers are instances of the same model. Two separate instances re-compiled the files in a clean environment (one also under the older pin), read every statement against its source (Eqs. (2)–(3) of the paper; upstream’s definitions line by line and by `rfl`), compiled instances showing that each hypothesis is satisfiable and counterexamples showing that the genericity and positive-semidefiniteness hypotheses cannot be dropped, and recorded their remarks; the verdicts are *verified* or *verified with remarks*, the reports and the answers are in the repository (`VERIFICATION_*.md`). This is not human review and no mathematician has read the files; the load-bearing evidence is kernel acceptance and the axiom lists. `TorusBound.lean` is conditional on a theorem that nobody here has checked.

6 Numerical tests on rusty-SUNDIALS

All gates were fixed before the runs they govern (`docs/designs/EXCITON_FLUID_PHASE1_PREPREG.md`, committed with independent reference numbers produced by a separate `numpy/mpmath` script before the Rust code existed); two amendments, with their reasons and dates, are recorded in the same file and below. Results are read as registered, failures included; exploratory runs (longer flows, a diagnostic trace) are labelled as such.

6.1 Instrument

rusty-SUNDIALS is the author’s own Rust implementation of the CVODE algorithms (<https://github.com/xaviercallens/rusty-SUNDIALS>); it is not the LLNL library of [5] and has not been reviewed by its authors. The code of this study is a small Rust crate on tree `996aaf0706e1` (tag `v11.6.0`, commit `6545abf`; the branch commit `5db8041` actually used has the same tree), using the `cvode` crate with the BDF method (orders 1–5), relative tolerance 10^{-10} and an analytic Jacobian; all production runs use BDF. Gate S0: the crate’s own tests pass (22 of 22, including the Adams-order and tight-tolerance tests); on $y' = -y$ to $t = 10$ the Adams method at $\text{rtol} = 10^{-10}$ needs 400 right-hand-side evaluations for a maximum relative error 9.76×10^{-10} (an older first-order build of the shared Python environment needed 2.2×10^6 evaluations on a similar test, CLAIM-109 of the ledger), and BDF at $\text{rtol} = 10^{-8}$ reaches 3.11×10^{-7} .

6.2 Known answers, and the first failure

The energy per particle of N points in a rectangular torus, with periodic images summed and the images of a point with itself included (as in the periodic extension the theorem is applied to), is $E = \frac{1}{2N} \sum_{(i,j,m) \neq (i,i,0)} g(|x_i - x_j + L \circ m|^2)$; the lattice energy is $e_{\text{lat}}(\rho) = \frac{1}{2} \sum_{a \neq 0} g(|a|^2)$. The gates were: (KA-1a) $\sum' r^{-3}$ over the unit-spacing triangular lattice against $6\zeta(3/2)L(3/2, \chi_{-3}) = 11.0341757349\dots$; (KA-1b) the Gaussian lattice energy against the Jacobi-theta closed form; (KA-1c) the lattice energies of all exactly summable kernels against an independent `numpy` sum; (KA-2) the Hartree integral $\int V d^2r = 4\pi d$ by quadrature; (KA-3) the Rust energy and force against an independent `numpy` implementation at 35 random configurations and the analytic Hessian against a finite difference of the force; (KA-4) the table of $e_{\text{H}}/e_{\text{lat}}$.

KA-1a failed as registered, first

The relative errors were 1.8×10^{-4} , 1.0×10^{-4} , 5.4×10^{-5} at cutoffs $R = 200, 400, 800$, halving with R . The lattice enumeration of both the `numpy` reference and the Rust code looped over $|m|, |k| \leq R/a + 3$ and missed the caps $|y| > \frac{\sqrt{3}}{2}R$ of the disk. Two implementations by the same author shared the slip; only the closed-form constant exposed it. Both were corrected, the references regenerated and the gates re-run unchanged (amendment A1). The exactly summable references changed by less than 2×10^{-16} , the sandwich table by at most 3×10^{-4} .

After the correction the gates read: KA-1a 1.35×10^{-8} at $R = 800$ (gate 10^{-6}); KA-1b 1.08×10^{-15} ; KA-1c 7.23×10^{-15} (and 2.25×10^{-13} at the reduced cutoffs of amendment A2); KA-2 8.06×10^{-13} ; KA-4 8.75×10^{-14} ; KA-3 energy 5.23×10^{-14} , force 6.39×10^{-16} (max-norm, relative), Hessian against finite differences 4.97×10^{-9} ; a Hessian with the sign flipped (planted) is detected.

6.3 The four-flavour flow

The imaginary-time flow $\dot{\psi}_i = -\psi_i \partial H / \partial n_i$, $n_i = \psi_i^2$, was integrated with CVODE (BDF, $\text{rtol} = 10^{-12}$) from 600 random positive starts at $B = 0.01, 0.04$ and 0.2 T. Every converged final state has its support inside one exchange pair (T1: no cross-pair supports among 600); the

polarisations and total densities of T2, T3 are reproduced to 1.68×10^{-12} (gate 10^{-9}); the field at which the grand potentials of the converged Π_A and Π_B states are equal is $B_c = 55.9$ mT, equal to the closed form to 1.57×10^{-12} (gate 10^{-8}); and starting from Π_A with a seed of 10^{-6} in flavour 3 (intervalley) the state leaves the intravalley pair at 0.730 T against the closed-form spinodal 0.7254 T (informational, within 5%). The large metastability window is a property of the mean-field model, not a prediction of the experiment. The grand potential decreases along every flow.

6.4 N-particle gradient flows

Design. For each (kernel, density, torus) batch, random starts (200 for K1, for K2 at $\rho = 0.5$ on T36c and for the control N1; 60 for all other K2–K4 batches under amendment A2, decided after K1 and K2 at $\rho = 0.5$ on T36c had run, which also lowered their cutoffs so that the neglected terms are of order $2 \times 10^{-13} e_{\text{lat}}$, KA-1c at those cutoffs) relax by $\dot{x} = -\nabla(NE)$ under CVODE BDF with the analytic Hessian as Jacobian, $\tau_{\text{max}} = 2000$; on commensurate tori, basin-hopping trials (up to 200, or 60) follow only if the lattice energy was not reached by the random starts (this happened only for K1 at $\rho = 1.5$ on T36c). Tori: T36c ($6a \times 3\sqrt{3}a$, commensurate), T64c ($8a \times 4\sqrt{3}a$, K1 only) and T36s (square, incommensurate, frustrated). Kernels: K1 e^{-t} ; K2 $e^{-\sqrt{t}}/\sqrt{t}$; K3 $t^{-3/2}e^{-0.02t}$; K4 the screened bilayer $2[t^{-1/2} - (t+1)^{-1/2}]e^{-0.02t}$ (all completely monotone); and the control N1 e^{-t^2} (not). Gates (registered): EX-1a no violation, $\min E/e_{\text{lat}} - 1 \geq -10^{-10}$; EX-1b attainment on commensurate tori, $\leq 10^{-9}$ (“not attained by this search” is not a failure of the inequality); EX-1c frustration on incommensurate tori; EX-2 the control reaches $E/e_{\text{lat}} \leq 0.99$; L2 the energy never increases by more than 10^{-9} relative; NEG planted bugs must make the gates fail.

kernel	ρ	torus	runs	conv.	$\min(E/e_{\text{lat}} - 1)$	$\# \leq 10^{-9}$	EX-1a	EX-1b / 1c	polished
K1	0.5	T36c	200	116	2.2×10^{-16}	90	pass	yes	–
K1	0.5	T36s	200	131	6.9×10^{-3}	0	pass	$+6.9 \times 10^{-3}$	6.9×10^{-3}
K1	0.5	T64c	200	47	2.4×10^{-15}	46	pass	yes	–
K1	1.5	T36c	400	0	5.9×10^{-9}	0	pass	no	1.1×10^{-15}
K1	1.5	T36s	200	0	7.9×10^{-7}	0	pass	$+7.9 \times 10^{-7}$	7.0×10^{-7}
K2	0.5	T36c	200	91	-2.0×10^{-14}	84	pass	yes	-4.4×10^{-15}
K2	0.5	T36s	60	38	3.0×10^{-3}	0	pass	$+3.0 \times 10^{-3}$	3.0×10^{-3}
K2	1.5	T36c	60	45	-1.1×10^{-13}	33	pass	yes	-8.6×10^{-14}
K2	1.5	T36s	60	33	1.3×10^{-3}	0	pass	$+1.3 \times 10^{-3}$	1.3×10^{-3}
K3	0.1	T36c	60	0	1.5×10^{-11}	5	pass	yes	-5.0×10^{-15}
K3	0.1	T36s	60	0	8.6×10^{-3}	0	pass	$+8.6 \times 10^{-3}$	8.6×10^{-3}
K3	0.5	T36c	60	39	-2.7×10^{-14}	27	pass	yes	-3.6×10^{-14}
K3	0.5	T36s	60	35	6.1×10^{-3}	0	pass	$+6.1 \times 10^{-3}$	6.1×10^{-3}
K4	0.1	T36c	60	0	5.9×10^{-10}	1	pass	yes	0
K4	0.1	T36s	60	0	7.6×10^{-3}	0	pass	$+7.6 \times 10^{-3}$	7.6×10^{-3}
K4	0.5	T36c	60	45	9.5×10^{-15}	27	pass	yes	–
K4	0.5	T36s	60	36	3.4×10^{-3}	0	pass	$+3.4 \times 10^{-3}$	3.4×10^{-3}
N1	1	T36c	200	200	-3.4×10^{-1}	–	ratio 0.663	–	–
N1	1	T36s	200	200	-3.3×10^{-1}	–	ratio 0.671	–	–
N1	4	T36c	200	200	-2.8×10^{-1}	–	ratio 0.723	–	–
N1	4	T36s	200	200	-2.8×10^{-1}	–	ratio 0.724	–	–

Table 3: N-particle results as registered (columns 1–9; “conv.” counts runs meeting the registered convergence criterion within τ_{max} ; every run counts in the gates with its final energy) and, in the last column, the *exploratory* result of a much longer flow from the best configuration of the case (not registered; “–”: not run). For the control N1, “EX-1a” shows the best ratio E/e_{lat} .

Results as registered. (i) **EX-1a passes** in all 17 completely monotone cases: the smallest $E/e_{\text{lat}} - 1$ over the 2060 runs of these cases is -1.1×10^{-13} , which is the roundoff and cutoff level of the lattice sums, not a violation. (ii) **EX-1b is attained** on 8 of the 9 commensurate cases; the exception, **K1 at $\rho = 1.5$ on T36c, failed as registered**: the best of 400 runs

is $+5.9 \times 10^{-9}$ and none of the 400 flows met the convergence criterion by $\tau_{\max} = 2000$. An exploratory continuation of the best configuration reaches $+1.1 \times 10^{-15}$ at $\tau \approx 4 \times 10^4$: the registered stopping rule is too short for the soft modes of a Gaussian kernel at that density, so “not attained by the registered search” is the accurate statement. (iii) **EX-1c**: the best energy found on every incommensurate torus lies above the lattice energy, by 7.9×10^{-7} (K1 at $\rho = 1.5$, strongly overlapping soft cores) to 8.6×10^{-3} (K3 at $\rho = 0.1$); the exploratory flows change these by at most 11 %. (iv) **EX-2 passes**: all 800 runs of the control end below the lattice energy, the best at $E/e_{\text{lat}} = 0.663$ to 0.724 ; the search did not find the best cluster crystal at $\rho = 4$ (a triangular cluster crystal gives 0.661 , the search 0.723), so it is not exhaustive. (v) **L2**: 0 violations in 2860 flows. (vi) **NEG**: a doubled lattice energy (the ordered-pair slip) makes EX-1a fail, a halved one makes EX-1b fail, and a Hessian with a sign flipped is detected by KA-3; the gates bite. Of the 2860 runs, 1404 did not meet the convergence criterion within $\tau_{\max}$ (they count with their final energy). On T36c, 40 to 55 percent of the random starts end within 10^{-9} of the lattice energy (90/200 for K1 at $\rho = 0.5$; 84/200 and 33/60 for K2 at $\rho = 0.5$ and 1.5 ; 27/60 for K3 and for K4 at $\rho = 0.5$), none for K1 at $\rho = 1.5$ within the registered flow length, and 5/60 and 1/60 for K3 and K4 at the low density $\rho = 0.1$ where the kernels are long-ranged and flat; 46/200 on T64c.

What this does and does not show. The energies are upper bounds on the minimum energy of N points on the torus; that they equal the lattice energy to roundoff on commensurate tori, and never go below it, is the behaviour the theorem predicts and is consistent with a correct implementation (two implementations by the same author and a closed-form constant agree, KA gates); on incommensurate tori nothing is predicted beyond positivity. It does not prove the theorem: a local search from random starts at $N \leq 64$ can only fail to find a counterexample.

6.5 Nonlocal-kernel mean field (Phase 2)

A second test (docs/designs/EXCITON_FLUID_PHASE2_PREREG.md, committed as 363475b with independent reference numbers before any run, with a control kernel of depth 40 that the run then did not use: see the end of this subsection) uses the mean-field functional $\Omega[\psi] = \int \frac{1}{2} |\nabla \psi|^2 + \frac{1}{2} n (U * n) - \mu n$ on a 16×16 periodic grid (256 unknowns, within CVODE’s dense Newton solver), with the bilayer kernel $\tilde{U}_B(k) = 4\pi(1 - e^{-kd})/k$ (kernel B, positive; units $\hbar = m = 1$, $d = 1$, $L = 16$, $\mu = 0.25$) and a control kernel C obtained by subtracting a Gaussian band of depth 28 and width 0.2 at $k_0 = 2\pi \cdot 3/L$ (not completely monotone). Results as registered:

- **P2-a, pass.** From 50 random starts the imaginary-time flow of kernel B reaches the uniform state in all runs (by $\tau = 100$, about 1100 BDF steps); worst $\max |n - n_0|/n_0 = 1.85 \times 10^{-14}$ and worst relative grand-potential error 1.57×10^{-15} .
- **P2-b, pass.** The solver’s own linearisation (central differences of the real-time right-hand side about the uniform state) reproduces the Bogoliubov spectrum $\omega_k^2 = \varepsilon_k(\varepsilon_k + 2n_0\tilde{U}(k))$ on all 255 nonzero wavenumbers to 1.15×10^{-10} (relative error of ω^2 ; 5.77×10^{-11} in ω) with real parts of the eigenvalues below 2.39×10^{-11} .
- **P2-c(i), pass.** For kernel C the same linearisation reproduces ω^2 to 6.33×10^{-9} and finds the same 16 unstable modes as the formula.
- **P2-d, pass.** A planted bug ($\tilde{U}(0)$ doubled) is caught on all 5 starts (density deviation 0.5).
- **P2-c(ii), failed as registered.** The flow of kernel C from uniform plus noise ended with a CVODE convergence failure in the first chunk, so the registered evidence (a modulation ≥ 0.1 and a grand potential below the uniform one by 10^{-3}) was not produced. A re-run with relaxed tolerances failed identically; a diagnostic trace in short chunks (exploratory) shows the grand potential decreasing monotonically while the modulation grows exponentially (rate ≈ 0.12), crossing 0.1 at $\tau \approx 38$, before the integration runs away at $\tau \approx 58$. The

reason is analytic, and it is a flaw of the control’s design: at fixed μ the functional of kernel C is *unbounded below* (a stripe of Gaussian profile has interaction energy $\frac{1}{2}\bar{n}^2 F(w)$ with $\min_w F = -19$; script `exploration/exciton/python/phase2_boundedness.py`), so no minimiser exists (the uniform state is an unstable stationary point); in this family a bounded, linearly unstable control does not exist at fixed μ . The failure therefore says nothing against the solver or the Lean statement, but the gate stays failed.

- **History of the control.** The registration committed as `363475b` had depth 40 (reference $\min \omega^2 = -0.42$). Within a minute, before any run, a boundedness check (a cosine profile only, which was itself incomplete, see above) led to depth 28: the reference numbers were regenerated and amendment A1 was written into the registration file, but A1 carries no date and was committed only with the results (`3203d78`). The control was therefore changed after registration and before the run, and the Phase 2 registration is weaker than that of Phase 1 in this respect.

This phase supports the consistency of `uniform_minimises` and of the Bogoliubov spectrum with the solver on the bilayer kernel and says nothing about excitons beyond that bare kernel (no screening, exchange, temperature or correlations).

7 Limitations and what we do not know

- **The theorem.** It is AI-generated, unrefereed, and was accepted by the Comparator tool in a run of the author’s own project on another machine; that check was not repeated here, and the adequacy of the statement has been audited only by AI agents (the same project’s, not independent). The same upstream release (7 October 2026) withdrew three other manuscripts of that body of AI-generated work (on Hodge classes and K3 surfaces) after a sign error was found; neither triangular manuscript was withdrawn. That calibrates the word “unrefereed”. Everything marked “conditional” above is conditional on the theorem. A finite- N numerical test can fail to falsify it; it cannot prove it.
- **Tightness.** In the experimental window the sandwich between the lattice and the Hartree energies is wide (Table 1: a factor 8–22); the ground-state energy lies between them and nothing here locates it. The lower bound is a structural statement and a solver test, not a prediction.
- **Kernels.** Complete monotonicity of the dual-gated device kernel is numerical evidence, not a proof; the Keldysh representation was checked numerically, not in Lean; positive definiteness of completely monotone kernels is classical (Schoenberg) and a hypothesis in Lean.
- **The mean-field test** (Phase 2) is a test of the solver and of `uniform_minimises` on a bare bilayer kernel, not of exciton physics (no screening, exchange, temperature, correlations); its control failed as registered, for a reason that is a flaw of the control’s design, and the control had been changed from its committed depth before the run.
- **N-particle tests** use $N = 36$ and 64, three tori, a local search from random starts, and (for K2–K4) the reduced budget of amendment A2; K5 and K6 were not run. The registered stopping rule is too short for the soft modes of the Gaussian kernel at $\rho = 1.5$ (EX-1b failed as registered there; exploratory longer flows attain the lattice energy to 10^{-15}).
- **The experiment.** Of Qi *et al.* only the arXiv version 1 text was read, plus the public abstract and Extended Data legends of the version of record (whose parameters differ); g_X and Δ are phenomenological; no experimental data were used; the four-flavour theorems are statements about the authors’ own model (the authors’ public code, which adds a phase term to the energy, was not used).
- **Lean.** Producer and verifiers are instances of one model; no human has read the files; two statements of `FourFlavour.lean` (T2, T3) are conditional on stationarity; the library’s Comparator runs predate the toolchain alignment.

- **The solver.** CVODE dense, ≤ 128 unknowns in the N-particle runs and 256 in the mean-field runs; no claim is made that it is the best of its kind, and no GPU or other code was measured.
- **Not contacted.** Neither the experimental group nor the authors of the theorem have seen this note.

Reproducibility

This note and its source archive are deposited on Zenodo (concept DOI [10.5281/zenodo.23289026](https://doi.org/10.5281/zenodo.23289026); this version [10.5281/zenodo.23289027](https://doi.org/10.5281/zenodo.23289027)). The repository state of the deposit is the git tag `exciton-note-1` of <https://github.com/xaviercallens/SocrateAI-Scientific-QuantumFluids> (directory `exploration/exciton/`, documents `docs/designs/EXCITON_FLUID_*.md`); the archive’s `MANIFEST.txt` records the commit and the SHA-256 of every file. The pre-registration of Phase 1 was committed as `d249544` before any run; that of Phase 2 as `363475b`, with the amendment history of Section 6.5 (its amendment A1 carries no date and was committed with the results). Environment: Rust 1.97.1; rusty-SUNDIALS tag `v11.6.0` (tree `996aaf0706e1`; the Cargo.toml path dependencies of the crate point to a clean clone at `~/xdev/rusty-SUNDIALS-c3`, to be edited); Python 3.12 with numpy 1.26.4 and mpmath 1.3.0; Lean 4.34.1 with Mathlib tag `v4.34.1` (`d13f23b7`; upstream `openai/math` commit `adc7f124`, definitions verified at `fd4aeeb2`); a 2013 laptop CPU (Intel i7-4930MX, 8 threads), shared and heavily loaded during the night runs. Commands: `cargo build --release` in `exploration/exciton/rust/qf-exciton-p1`; `ex1`, `ka`, `s0`, `ff`, `polish`, `mf` (with `QF_AMEND=A2` for the amended cases); `phase1_summarise.py`, `phase2_summarise.py` and `phase1_paper_numbers.py` in `exploration/exciton/python/` regenerate the numbers of Tables 1 and 3, of Sections 6.2–6.5 and of the abstract from the run records; the Keldysh identity, the dual-gate derivative signs and the e_H/e_{lat} values with a short cutoff come from the scratch scripts copied unmodified into `exploration/exciton/python/study/`, the boundedness numbers from `phase2_boundedness.py`, the critical fields from `critical_field_versions.py`. The field `min_dist` of the Phase 1 run records is a placeholder (always 0). Lean: `lake env lean` on each file of `exploration/exciton/lean/`, one at a time. Random numbers are SplitMix64 with the seeds stated in the pre-registration. The note belongs to the programme of the QuantumFluids library [12], its book [13] and its software paper [14].

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